AI for Cyber Security

What AI-assisted coding does for small teams

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The route

01	Intro	00–08
02	Guardrails	08–16
03	Hands-on setup	16–42
04	Examples and demos	42–70
05	Skills, agents, and plugins	70–85
06	Application builds	85–110
07	Getting started	110–115
08	Close	115–120

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Who this is for

> **You, if your “security team” is also the help desk, the network team, and the person who resets passwords.**

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What AI-assisted coding is



Autocomplete

Finishes the line you started.



Chat

You ask, copy, run, and check.



Agents

Read files, write code, run it, and respond to results.

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The work nobody had time to script

Log triage

Review the outliers.

Config review

Check network and identity exports.

Glue scripts

Connect the systems you already run.

Incident prep

Draft playbooks and evidence collection.

Vendor code

Read it before running as admin.

Compliance

Draft requests and control evidence.

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Why small teams gain the most



A large security team

Already has scripting and analysis specialists. The tools can help them move faster.



A small district IT team

Can tackle useful work that previously stayed in the backlog.

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The caveat before we go on

> **It will confidently write a script that deletes the wrong accounts. You still have to read it.**

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02

Guardrails

Five rules before any demo.

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The five guardrails

- Read before you run.
- Work in a safe place.
- Protect the data.
- Keep permissions tight.
- Verify the output.

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Read it. Give it a safe place to work.



Read before you run

Inspect actions that delete, disable, write, or send.



Work in a safe place

Use a scratch folder, test tenant, or non-production copy.

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Protect the data and the access

Protect the data

No student records, passwords, API keys, or unapproved district information in prompts.

Keep permissions tight

Use the narrowest access required. Review plugins and shell commands.

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Confidence is not evidence

> **Run the query against known data. Test the script. Spot-check the citations.**

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03

Hands-on setup

Claude Code running, a project on GitHub, and an editor ready.

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Get Git on Mac or PC

github.com/OCDEkr/Cite-Talk-2026

MAC · TERMINAL

xcode-select --install
Finish the installer; reopen Terminal.
With Homebrew already installed: brew install git

PC · GIT BASH

git-scm.com/install/windows
Install Git for Windows, including Git Bash.
Open Git Bash from the Start menu.

Install Git first. “git clone” is the command that downloads the repository.

Clone the talk and open the guide

```
git --version
cd ~/Desktop
git clone https://github.com/OCDEkr/Cite-Talk-2026.git
cd Cite-Talk-2026
```

```
Open Cite Talk Guide.html in your browser.
Use Setup Guide.md and the Cheat Sheets folder alongside it.
```

github.com/OCDEkr/Cite-Talk-2026

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Check Git, Python, and Node

MAC · TERMINAL

```
brew install git python node

git --version
python3 --version
node --version
```

PC · POWERSHELL, THEN GIT BASH

```
winget install Python.Python.3.12
winget install OpenJS.NodeJS.LTS

# Reopen Git Bash, then check:
python --version
node --version
```

Install Claude Code, then sign in

MAC / LINUX / WSL

```
curl -fsSL https://claude.ai/install.sh |
bash

claude --version
claude doctor
claude
```

WINDOWS · POWERSHELL

```
irm https://claude.ai/install.ps1 | iex

# Reopen Git Bash:
claude --version
claude doctor
claude
```

Inside Claude Code: /status

Permissions and the workshop shortcut

```
# Bash · Windows Git Bash or Linux
echo 'alias clauded="claude --dangerously-skip-permissions"' >> ~/.bashrc
. ~/.bashrc
```

```
# Zsh · macOS default shell
echo 'alias clauded="claude --dangerously-skip-permissions"' >> ~/.zshrc
. ~/.zshrc
```

Choose your shell. >> appends; the leading dot reloads the file. clauded skips permission checks.

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Connect your own GitHub account

```
ssh-keygen -t ed25519 -C "you@yourdistrict.org"
cat ~/.ssh/id_ed25519.pub
ssh -T git@github.com
```

```
GitHub → Settings → SSH and GPG keys → New SSH key
```

VS Code and the GitHub CLI



VS Code

Open the project. Install Python and Claude Code extensions; WSL users add the WSL extension.



GitHub CLI

Install gh, then run gh auth login. Check with gh auth status.

Connect your first project

Set up this folder as a git repository named "my-project", create a private GitHub repository under the username YOUR_GITHUB_USERNAME, add it as the origin remote using SSH, and push. Use the gh command for anything that needs GitHub itself.

mkdir -p ~/my-project && cd ~/my-project → claude → /init

Pause: finish your setup

12 MINUTES · WORK, THEN CHECK

DO	Follow the setup steps in Cite Talk Guide.html. Finish login, editor, and your first project.
CHECK	Claude answers in the right folder. Your own private repository shows the project files.
IF BLOCKED	Use the guide’s troubleshooting steps or pair with a working machine.

Optional ways to extend your setup



No paid plan?

Step 4b describes OpenRouter. Check provider availability and data terms before use.



Optional tools

herdr, Codex, and Hermes have their own setup sections. Everyone uses Claude Code in this class.

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04

Examples and demos

GRC, red team, blue team, code review, and OSINT.

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Five areas. Useful outputs.

- GRC: policies, procedures, and vendor vetting.
- Red team: exposure reports, awareness, and tabletops.
- Blue team: log triage, detection rules, and alert explanations.
- Code review: vendor scripts and your own automation.
- OSINT: your own organization’s public exposure.

GRC: turn documents into useful work

Policy gaps

Compare an AUP with CIS IG1 and NIST CSF Govern.

Policy to procedure

Turn the 24-hour offboarding requirement into an auditable checklist.

Vendor vetting

Use provided terms and privacy documents. Mark silence “not addressed”.

GRC: turn a policy into a procedure

Our policy says: "User accounts must be disabled within 24 hours of employee separation and student withdrawal." We use Entra ID, Google Workspace, Aeries SIS, and a Fortinet VPN. HR notifies IT by email. Write a step-by-step offboarding procedure that satisfies this policy, with a step for each system, who does it, how it is verified, and what evidence to keep for an audit. Format it as a checklist we can paste into a ticket template.

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Pause: make the checklist usable

5 MINUTES · WORK, THEN CHECK

DO Run the offboarding prompt. Ask a follow-up that makes the checklist useful to your team.

CHECK Find the role, verification, and evidence for each named system.

DISCUSS Which steps came from the policy, and which still need a district decision?

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Red team: make sense of your scan

Read scan-report.pdf. This is an external exposure report for assets our district owns. Summarize it for a non-technical cabinet: what is exposed, why it matters in plain language, and the business risk if ignored. Then produce a remediation table for my team, sorted by risk, with the specific fix, the system owner who would do it, and a rough effort estimate. Flag anything that looks like a false positive and say why.

Existing file: Files/scan-report.pdf

Pause: find Monday’s priorities

5 MINUTES · WORK, THEN CHECK

DO Run the exposure-report prompt against Files/scan-report.pdf.

CHECK Trace one recommended fix back to the report’s finding and evidence.

DISCUSS Would cabinet understand the summary? What needs confirmation before acting?

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Blue team: triage a log export

I have an Entra ID sign-in log export in signins.csv. Write and run a script that flags anything worth a human look: successful logins from new or unusual countries, impossible travel between two logins, repeated failures followed by a success, and logins outside normal hours for that account. Show me a table of the flagged events with the reason each was flagged, sorted by how suspicious. Do not change the file.

Demo when a suitable signins.csv is staged; review the code and the evidence.

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Code review: read it before it runs

Read deploy.ps1. I am about to run this as a domain admin. Walk me through what it actually does, line by line at a high level. Flag anything that reaches out to the internet, changes permissions, touches the registry, deletes files, or hides its own activity. Tell me the three things most likely to cause damage if I run it as-is, and what I should verify before running it.

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OSINT: see what an outsider sees

Using only public web sources, build a profile of what an outsider could easily learn about Example Unified School District that would help them target us: our email address format, names and titles of IT and finance staff who appear on the public site, technologies mentioned in job postings, and any documents indexed publicly that mention internal systems. Present it as a risk briefing with what to remove or lock down for each item. Do not attempt to access anything non-public.

Five-minute break

> **Save your work. Return at elapsed minute 70.**

Next: skills, agents, plugins, then the app build.

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05

Skills, agents, and plugins

A skill for a task. An agent for a job. A plugin to package them.

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Smallest to largest



Skill

Instructions Claude loads for a repeatable task.



Agent

A specialist with its own context and selected tools.



Plugin

Skills, agents, hooks, and connectors packaged for installation.

Build a skill: a folder and a file

```
~/.claude/skills/config-review/SKILL.md
```

```
---
description: Reviews a network device config against
district security standards. Use for config reviews.
---
Check weak credentials, SNMP communities, telnet,
management access, and missing syslog.
Return issue, source line, severity, and proposed fix.
Do not change the configuration.
```

Pause: try your config-review skill

4 MINUTES · WORK, THEN CHECK

DO Create the skill from the tutorial. Invoke /config-review with the supplied Cisco configuration.

CHECK Trace one finding to its exact source line.

DISCUSS Which findings are proven by the file, and which require network context?

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Build an agent: a narrow job

```
~/.claude/agents/log-triage.md
```

```
---
name: log-triage
description: Reviews exported security logs.
tools: Read, Bash, Grep
model: sonnet
---
Find anomalies and return ranked evidence.
Never modify the source file.
```

Bash can change files. Tool restrictions and instructions are different controls.

Package the pieces as a plugin

DIRECTORY

```
my-security-pack/
.claude-plugin/plugin.json
skills/config-review/SKILL.md
agents/log-triage.md
```

MANIFEST

```
{
"name": "my-security-pack",
"version": "0.1.0",
"description": "K-12 security pack"
}
```

Install ECC, then inspect what it adds

```
/plugin marketplace add https://github.com/affaan-m/ECC
/plugin install ecc@ecc
```

```
/plugin
```


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Five focused plugins

- Humanizer: edit AI-sounding prose.
- Superpowers: plan, test, and debug systematically.
- Frontend design: build the interface.
- Impeccable: audit and improve the front end.
- TDD: use a test-first workflow.

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06

Application builds

From one-off scripts to a working application.

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A web-based Zenmap clone

I would like to build a user friendly interface to nmap similar to zenmap but a web app. Do your own research while planning and ask me any clarifying questions.

Fresh project folder → /model opus → send the original prompt

Have answers ready



Scan types

Discovery, ports, and service/version information. Leave OS detection off initially.



Stack and users

Simple local backend and web front end. One local user.



Inputs and targets

Real scans on authorized targets, plus saved Nmap XML import.

Pause: build it together

15 MINUTES · WORK, THEN CHECK

PLAN Answer the questions and review the plan before code is written.

BUILD Let Claude create the local front end and the backend that calls Nmap.

RUN Launch the app. Run one authorized scan and inspect the actual result.

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Verify the app you just built

- A real scan runs through the browser and backend.
- Raw output and host/port/service views agree.
- Saved Nmap XML imports into the result views.
- Use the fictitious 50-machine scan to check display and navigation.
- Explain an error, a validation rule, and the next improvement.

07

Getting started

Pick a plan. Choose a first task. Build the habit.

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Pick a plan that fits the work

Free	$0	Claude chat; alternative-provider setup is in the guide.
Pro	$20/mo	About $17/mo with annual billing; includes Claude Code.
Max / Ultra	$100 / $200/mo	Higher individual usage tiers.
Team / Enterprise	Per-seat / quoted	Shared administration and district adoption.
API	Usage-based	Metered automation and scripts.

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Choose your first project

Low stakes

Read data or work on a copy in a scratch folder.

Repetitive

Choose a task you do often enough to reuse the result.

Checkable

Know what a correct answer looks like.

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Build the habit. Bring it to your team.

One repeatable task

Write a skill and measure the time spent and corrections made.

One small pilot

Show evidence, agree on data/account rules, then scale deliberately.

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08

Close

One frame. One action. Your questions.

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A fast junior analyst

> **Capable, tireless, and never trusted blind.**

Monday: pick one annoying task. Give it one hour.

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Take the material with you

- Cite Talk Guide.html: setup, prompts, guardrails, and resources.
- Setup Guide.md: the step-by-step installation reference.
- Guardrails Checklist.md: the habits to keep by your desk.
- Resource Index.md and Cheat Sheets: more tools when you need them.

github.com/OCDEkr/Cite-Talk-2026

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Thank you. What would you want it to do first?

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